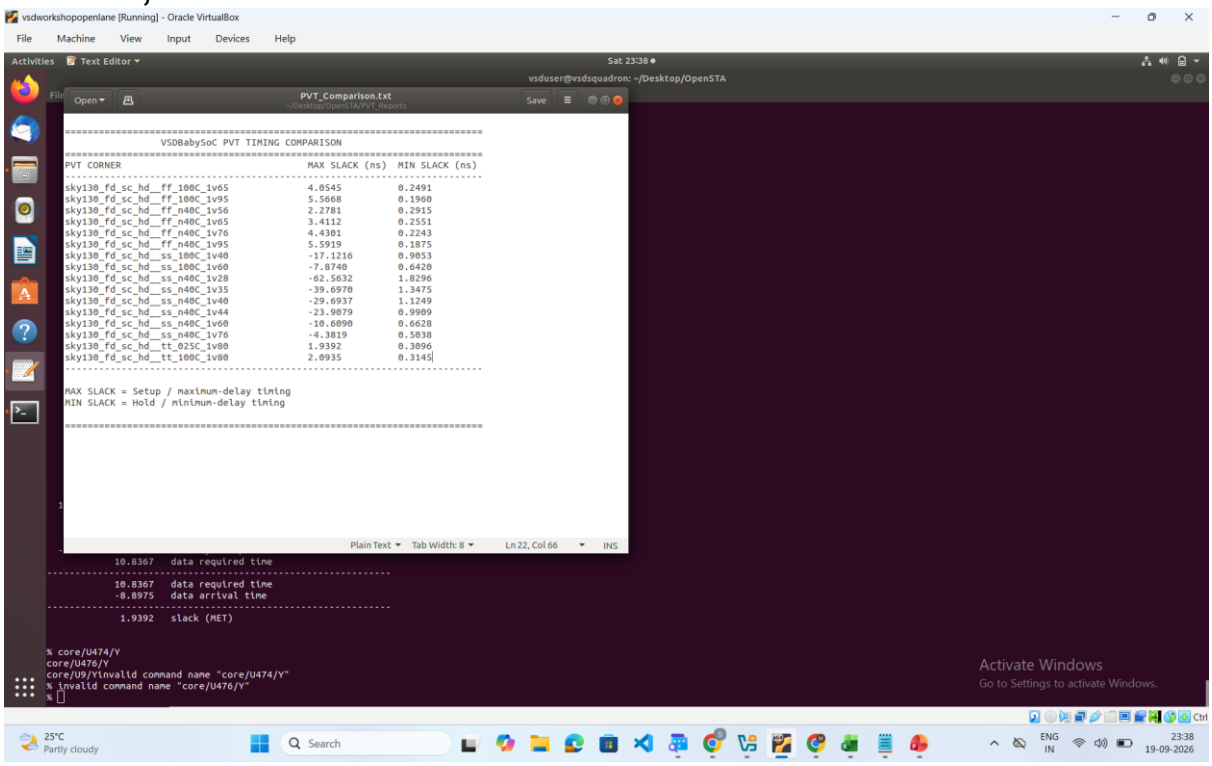
Static Timing Analysis

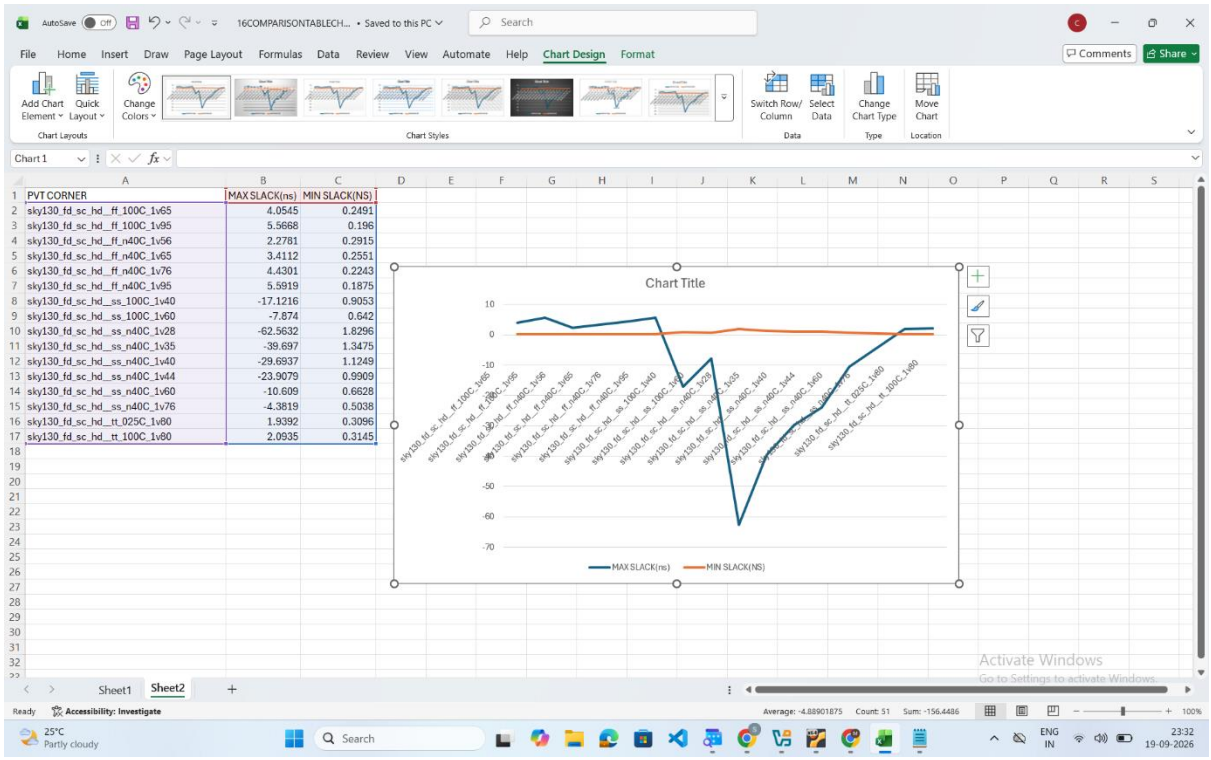
1. PVT Corner Overview

PVT means Process, Voltage and Temperature. The timing libraries represent different combinations of manufacturing process, supply voltage and temperature. TT denotes typical-typical process, FF fast-fast process, and SS slow-slow process. Lower-voltage corners generally represent reduced supply voltage conditions, while the temperature is encoded in the corner name. Corner naming example: sky130_fd_sc_hd__ss_n40C_1v28 → SS process, −40°C, 1.28 V.



2. Timing Comparison Graph

Positive slack indicates the reported timing check is met; negative slack indicates a timing violation for that check.



PVT CORNER	MAX SLACK(ns)	MIN SLACK(NS)
sky130_fd_sc_hd__ff_100C_1v65	4.0545	0.2491
sky130_fd_sc_hd__ff_100C_1v95	5.5668	0.196
sky130_fd_sc_hd__ff_n40C_1v56	2.2781	0.2915
sky130_fd_sc_hd__ff_n40C_1v65	3.4112	0.2551
sky130_fd_sc_hd__ff_n40C_1v76	4.4301	0.2243
sky130_fd_sc_hd__ff_n40C_1v95	5.5919	0.1875
sky130_fd_sc_hd__ss_100C_1v40	-17.1216	0.9053
sky130_fd_sc_hd__ss_100C_1v60	-7.874	0.642
sky130_fd_sc_hd__ss_n40C_1v28	-62.5632	1.8296
sky130_fd_sc_hd__ss_n40C_1v35	-39.697	1.3475
sky130_fd_sc_hd__ss_n40C_1v40	-29.6937	1.1249
sky130_fd_sc_hd__ss_n40C_1v44	-23.9079	0.9909
sky130_fd_sc_hd__ss_n40C_1v60	-10.609	0.6628
sky130_fd_sc_hd__ss_n40C_1v76	-4.3819	0.5038
sky130_fd_sc_hd__tt_025C_1v80	1.9392	0.3096
sky130_fd_sc_hd__tt_100C_1v80	2.0935	0.3145

3. STA Method and Commands

The STA flow used the synthesized VSDBabySoC netlist, the selected Sky130 HD standard-cell timing library, the AVSDPLL and AVSDDAC Liberty models, and the VSDBabySoC SDC constraints. read_liberty read_liberty src/lib/avsdpll.lib read_liberty src/lib/avsddac.lib read_verilog src/vsdbabysoc.synth.v link_design vsdbabysoc read_sdc src/sdc/vsdbabysoc_synthesis.sdc report_checks -path_delay min_max -format full_clock_expanded -digits 4

4. Detailed Timing Output — TT 25°C, 1.80 V

Worst minimum/hold path: core/CPU_imm_a2_reg[10] → core/CPU_imm_a3_reg[10], slack = 0.3096 ns (MET). Worst maximum/setup path: core/CPU_is_addi_a3_reg → core/CPU_Xreg_value_a4_reg[27][30], slack = 1.9392 ns (MET).

Representative net report for core/U474/Y Net core/n144
Pin capacitance: 0.00-0.00
Wire capacitance: 0.00
Total capacitance: 0.00-0.00
Number of drivers: 1 Number of loads: 2 Number of pins: 3 Driver pins core/U474/Y output (sky130_fd_sc_hd__nor2_1) Load pins core/U229/A input (sky130_fd_sc_hd__nand2_1) core/U476/A input (sky130_fd_sc_hd__nand2_1) Representative slew reports core/U474/Y ^ 0.11:0.11 v 0.03:0.04 core/CPU_is_addi_a3_reg/Q ^ 0.07:0.07 v 0.04:0.04 core/CPU_Xreg_value_a4_reg[27][30]/D ^ 0.11:0.17 v 0.03:0.14

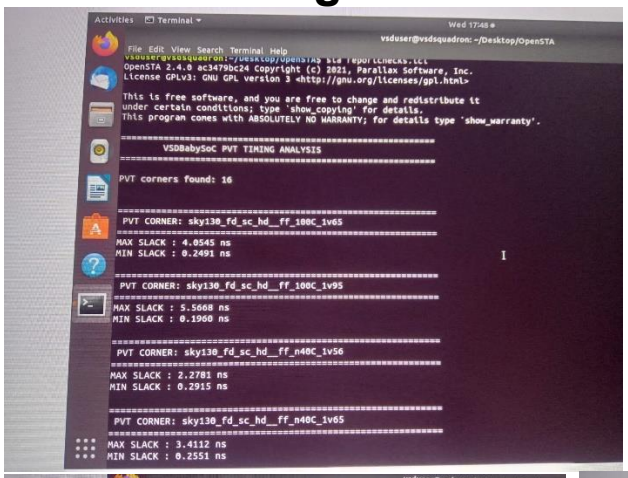
5. Corner-by-Corner Interpretation

ff_100C_1v65 — Process: FF; Temperature: 100°C; Voltage: 1.65 V; Max slack: 4.0545 ns; Min slack: 0.2491 ns.
ff_100C_1v95 — Process: FF; Temperature: 100°C; Voltage: 1.95 V; Max slack: 5.5668 ns; Min slack: 0.1960 ns.
ff_n40C_1v56 — Process: FF; Temperature: -40°C; Voltage: 1.56 V; Max slack: 2.2781 ns; Min slack: 0.2915 ns.
ff_n40C_1v65 — Process: FF; Temperature: -40°C; Voltage: 1.65 V; Max slack: 3.4112 ns; Min slack: 0.2551 ns.
ff_n40C_1v76 — Process: FF; Temperature: -40°C; Voltage: 1.76 V; Max slack: 4.4301 ns; Min slack: 0.2243 ns.
ff_n40C_1v95 — Process: FF; Temperature: -40°C; Voltage: 1.95 V; Max slack: 5.5919 ns; Min slack: 0.1875 ns.
ss_100C_1v40 — Process: SS; Temperature: 100°C; Voltage: 1.40 V; Max slack: -17.1216 ns; Min slack: 0.9053 ns.
ss_100C_1v60 — Process: SS; Temperature: 100°C; Voltage: 1.60 V; Max slack: -7.8740 ns; Min slack: 0.6420 ns.
ss_n40C_1v28 — Process: SS; Temperature: -40°C; Voltage: 1.28 V; Max slack: -62.5632 ns; Min slack: 1.8296 ns.

ss_n40C_1v35 — Process: SS; Temperature: -40°C; Voltage: 1.35 V; Max slack: -39.6970 ns; Min slack: 1.3475 ns.

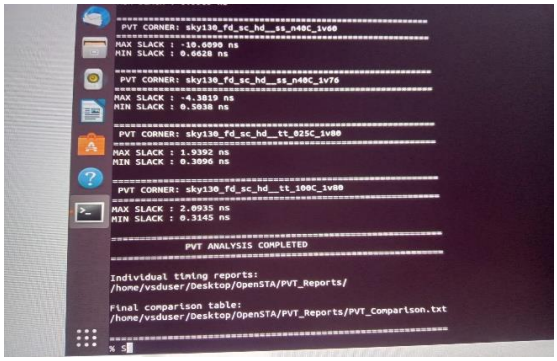
ss_n40C_1v40 — Process: SS; Temperature: -40°C; Voltage: 1.40 V; Max slack: -29.6937 ns; Min slack: 1.1249 ns.
ss_n40C_1v44 — Process: SS; Temperature: -40°C; Voltage: 1.44 V; Max slack: -23.9079 ns; Min slack: 0.9909 ns.
ss_n40C_1v60 — Process: SS; Temperature: -40°C; Voltage: 1.60 V; Max slack: -10.6090 ns; Min slack: 0.6628 ns.
ss_n40C_1v76 — Process: SS; Temperature: -40°C; Voltage: 1.76 V; Max slack: -4.3819 ns; Min slack: 0.5038 ns.
tt_025C_1v80 — Process: TT; Temperature: 025°C; Voltage: 1.80 V; Max slack: 1.9392 ns; Min slack: 0.3096 ns.
tt_100C_1v80 — Process: TT; Temperature: 100°C; Voltage: 1.80 V; Max slack: 2.0935 ns; Min slack: 0.3145 ns.

6.Terminal Logs



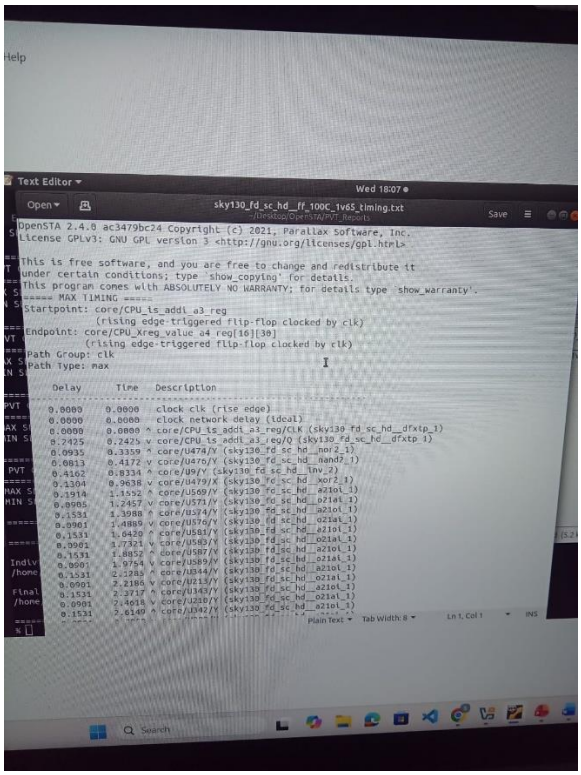


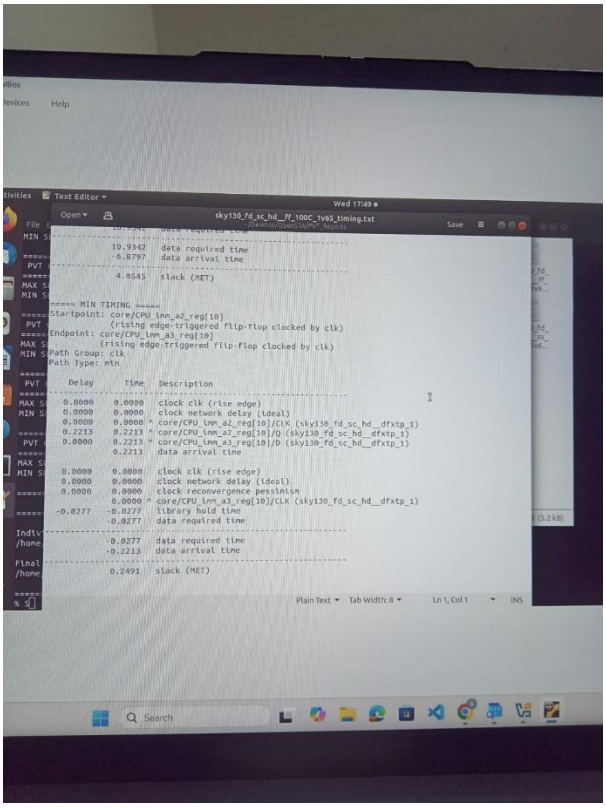




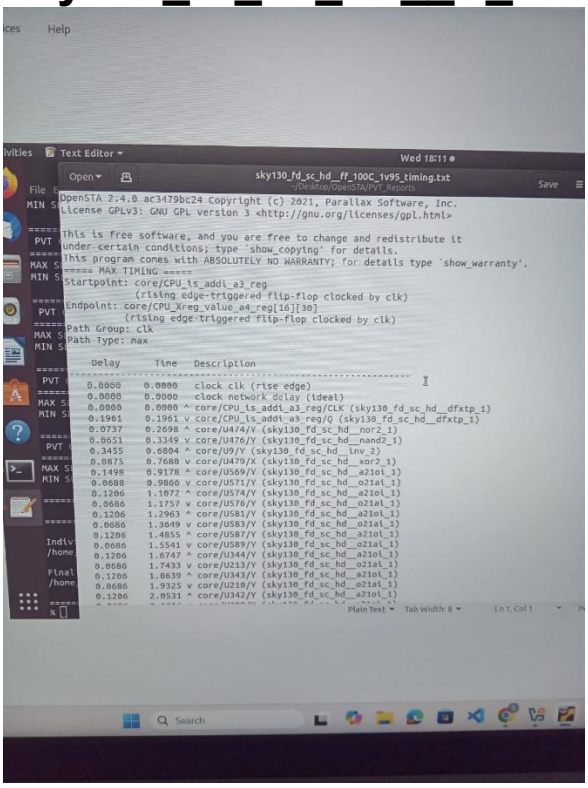
7. Individual timing reports

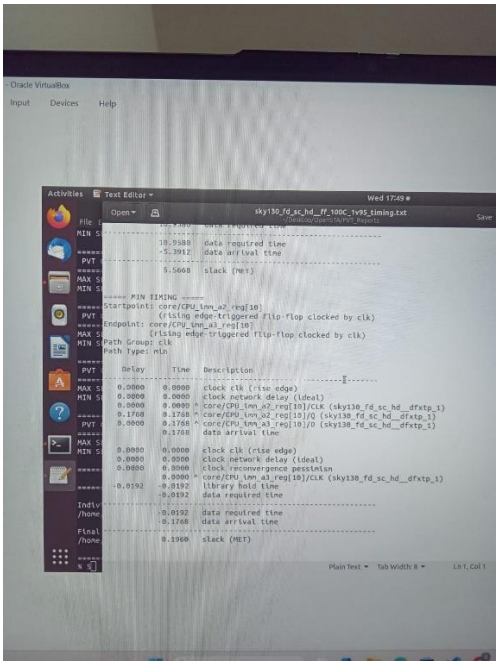
- **sky130_fd_sc_hd__ff_100C_1v65**



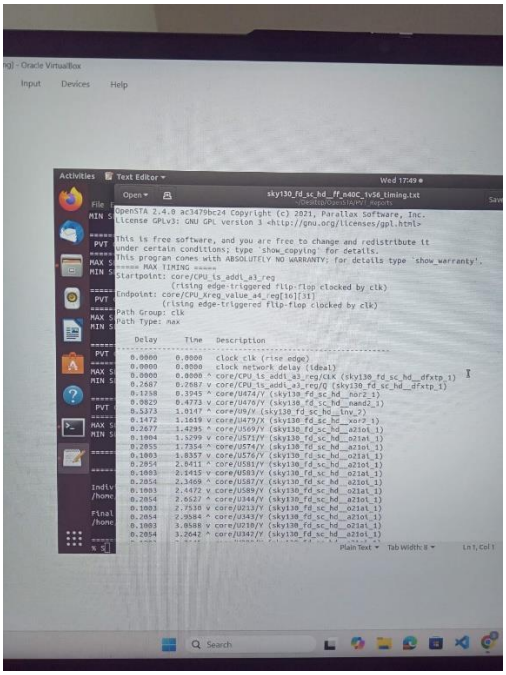


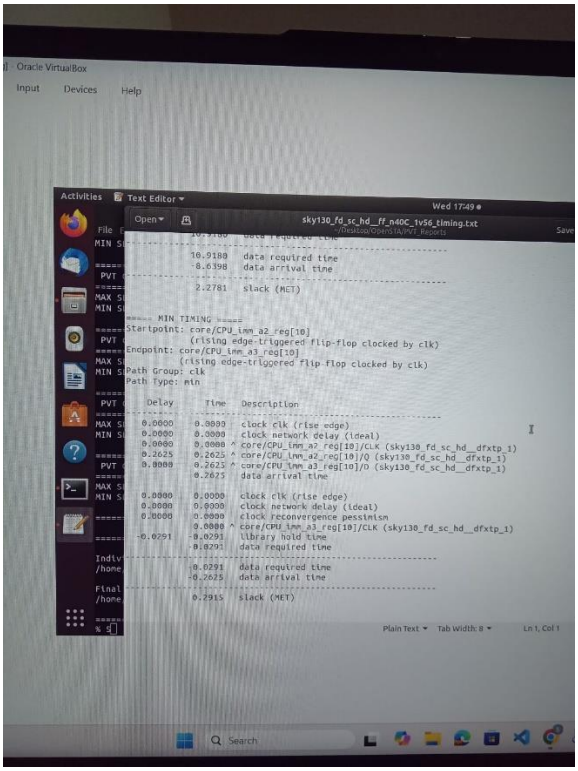
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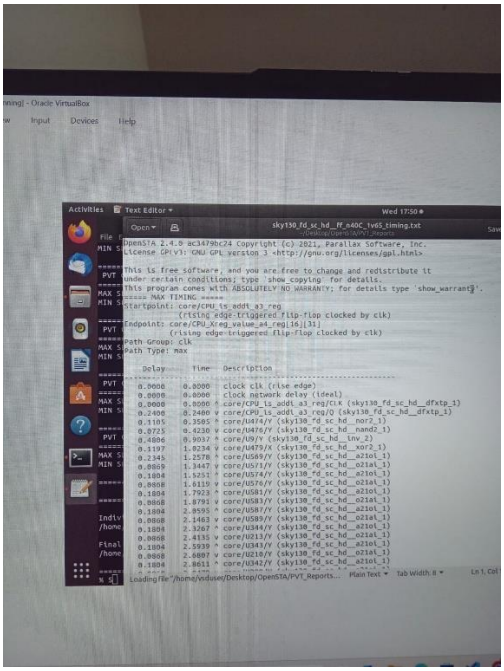


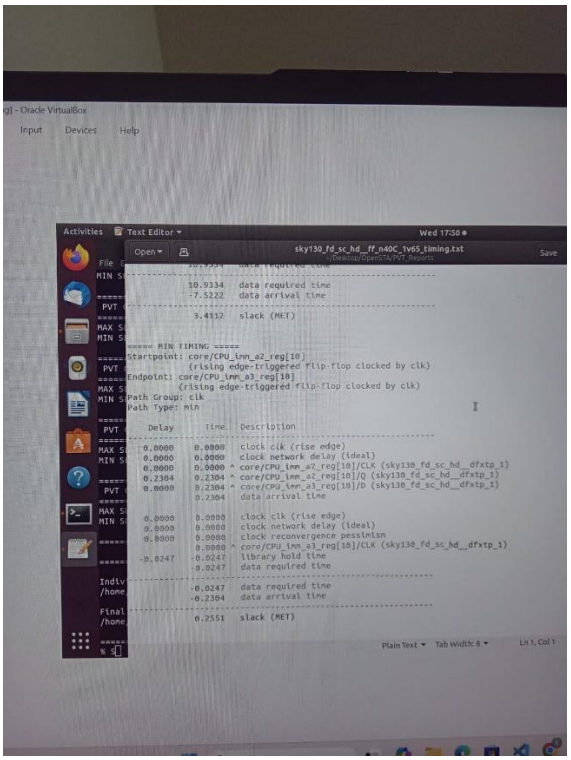
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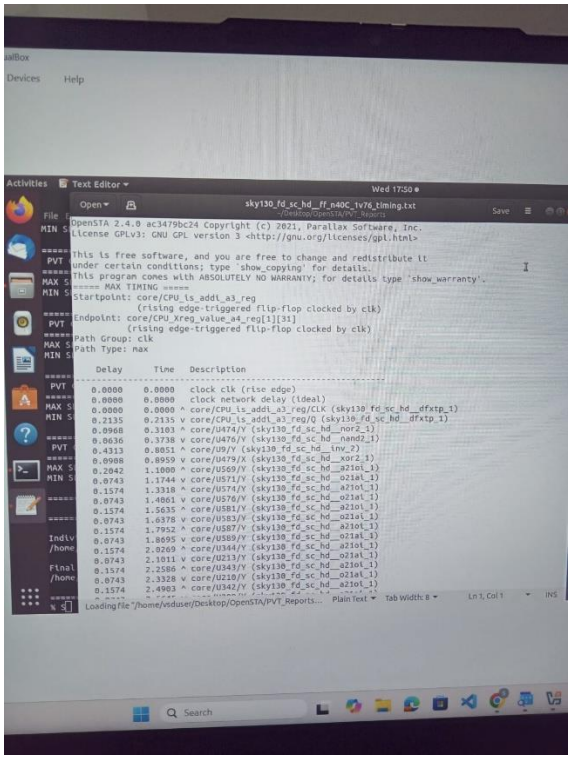


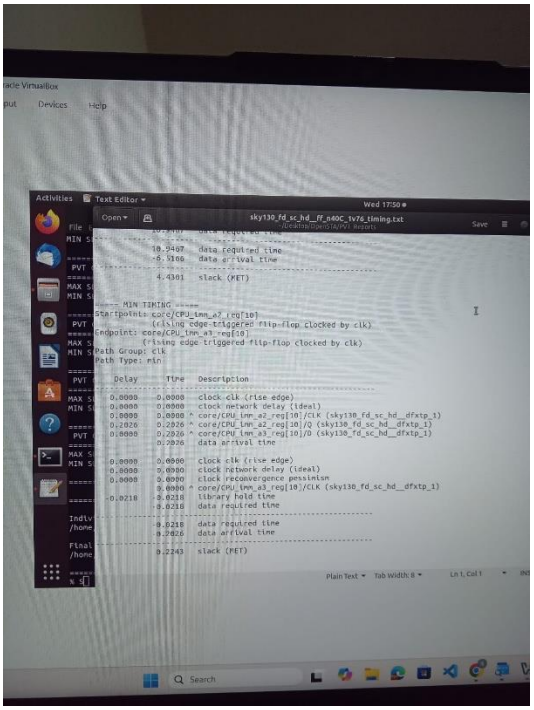
- **sky130_fd_sc_hd__ff_n40C_1v65**



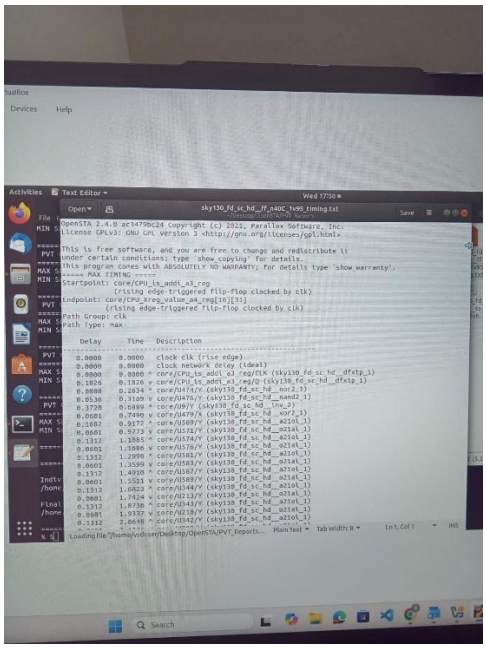


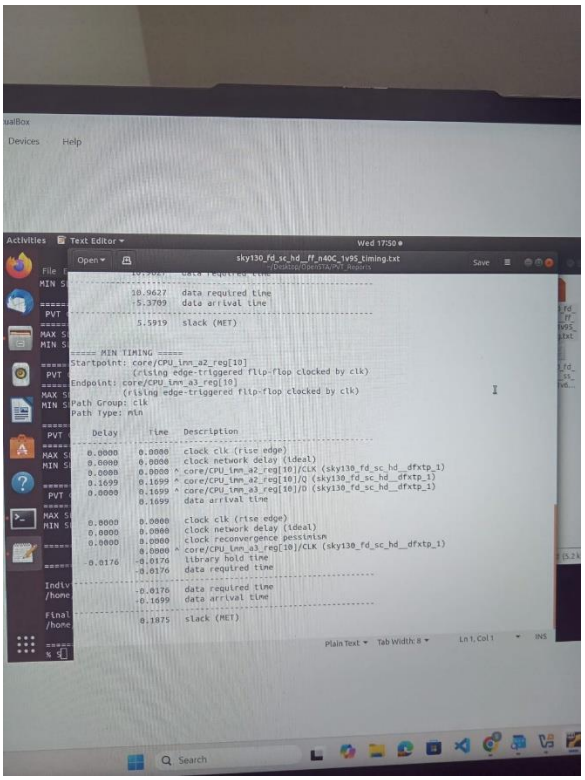
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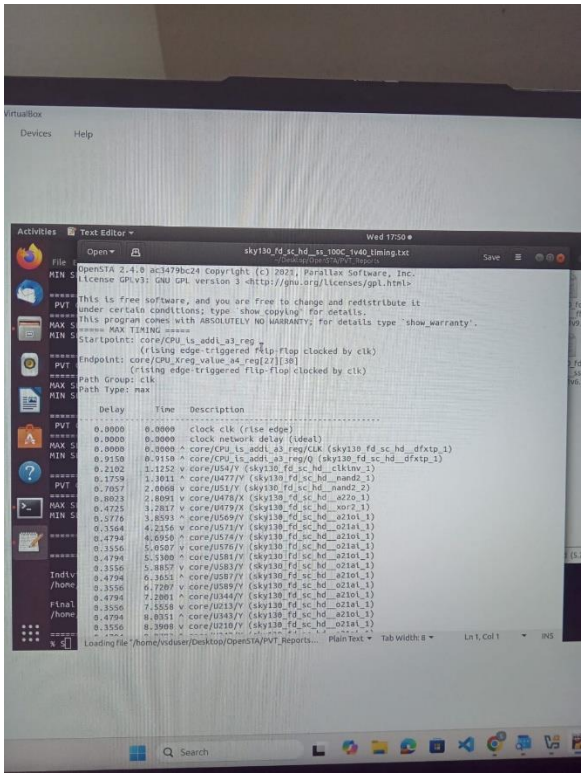


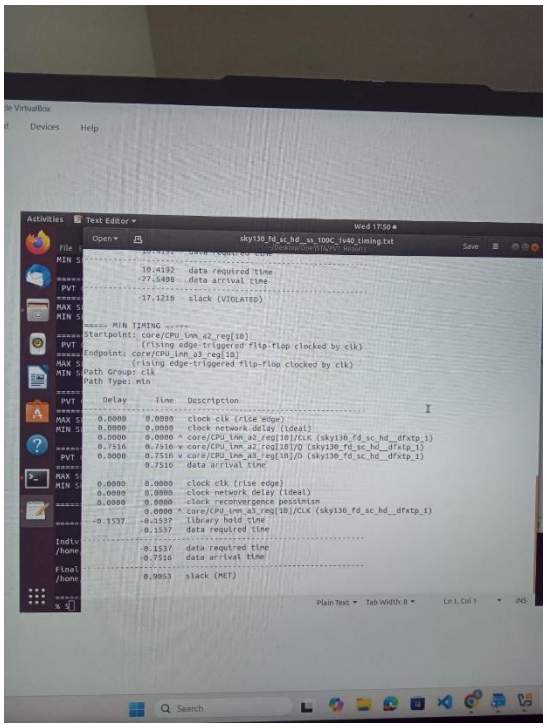
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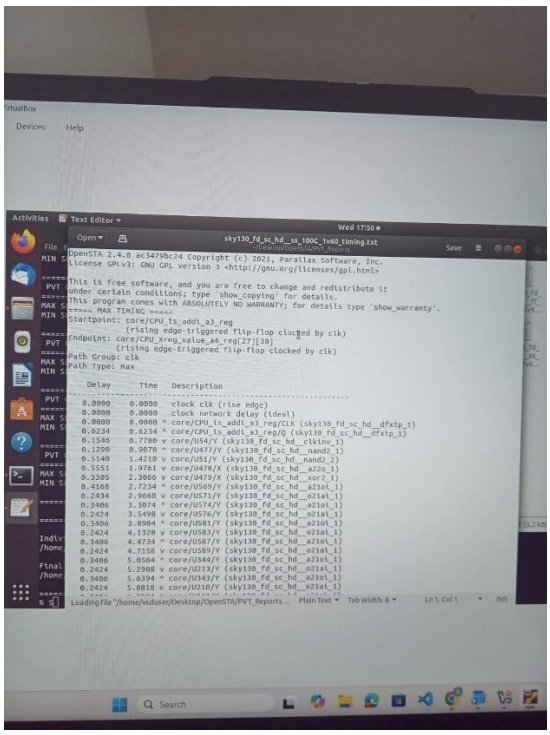


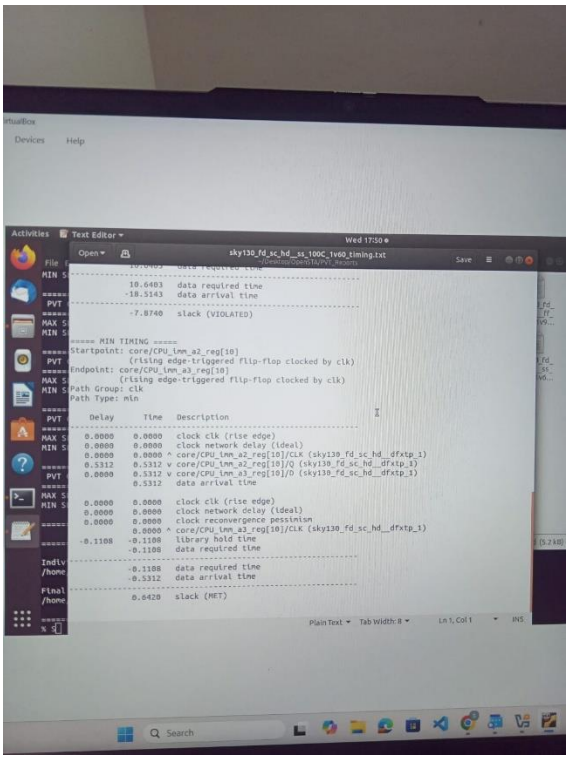
- **sky130_fd_sc_hd__ss_100C_1v40**



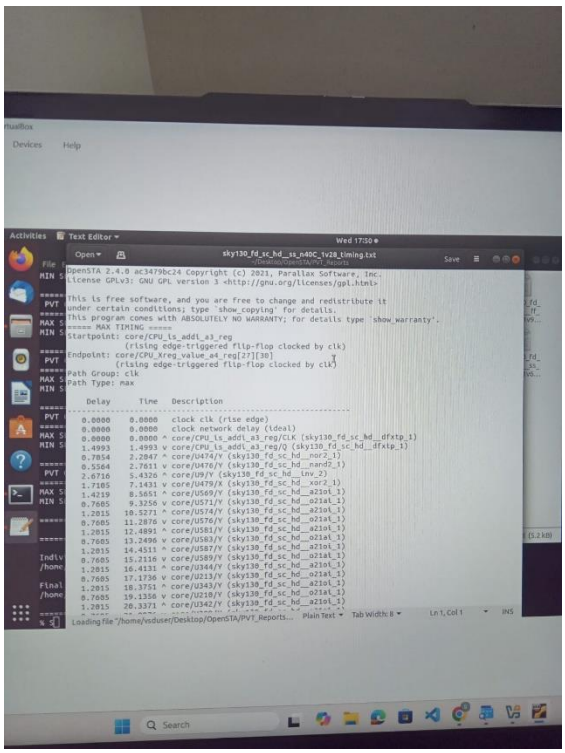


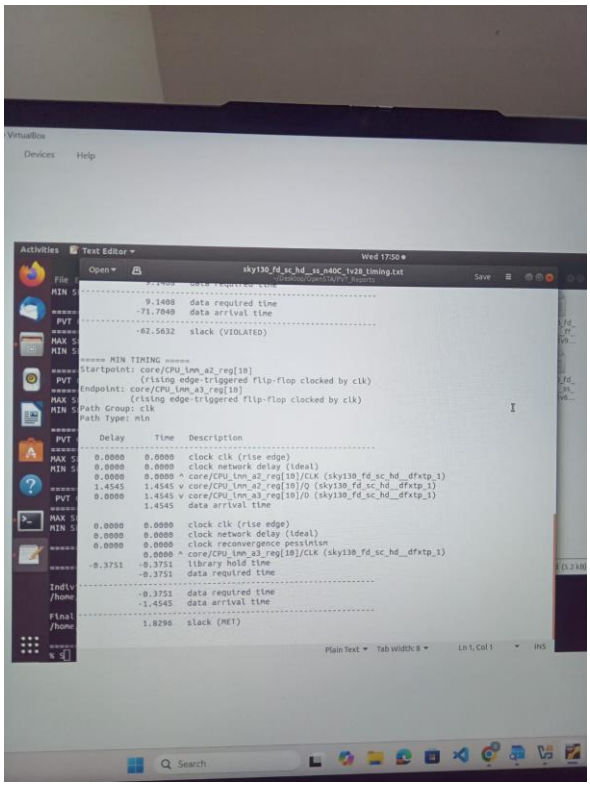
- **sky130_fd_sc_hd__ss_100C_1v60**



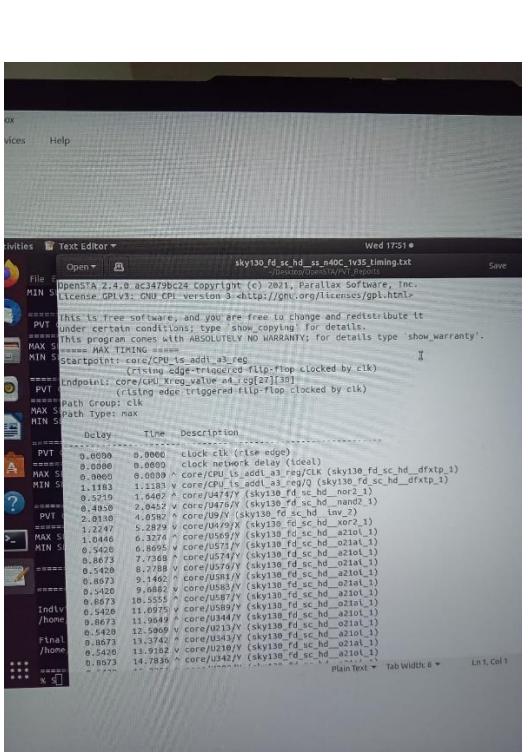


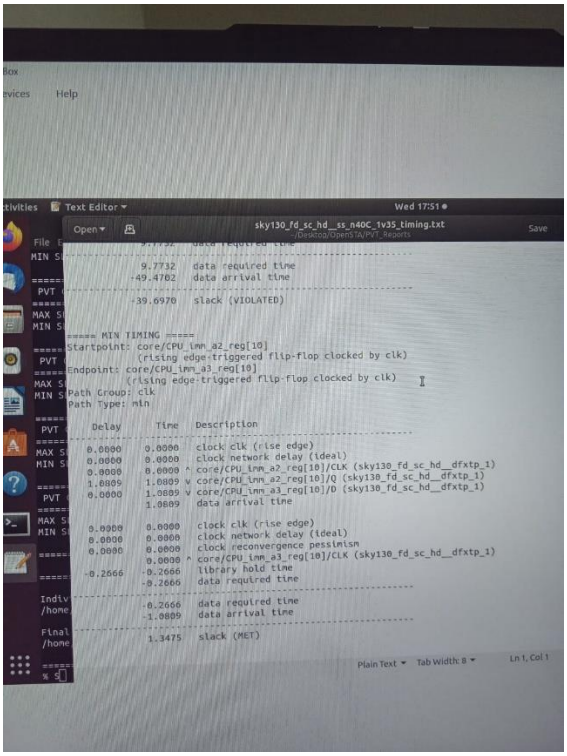
- **sky130_fd_sc_hd__ss_n40C_1v28**



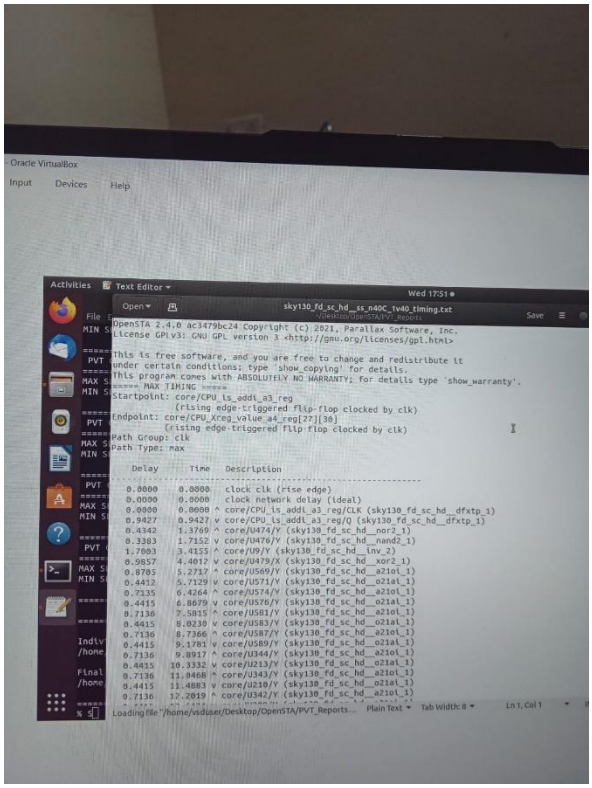


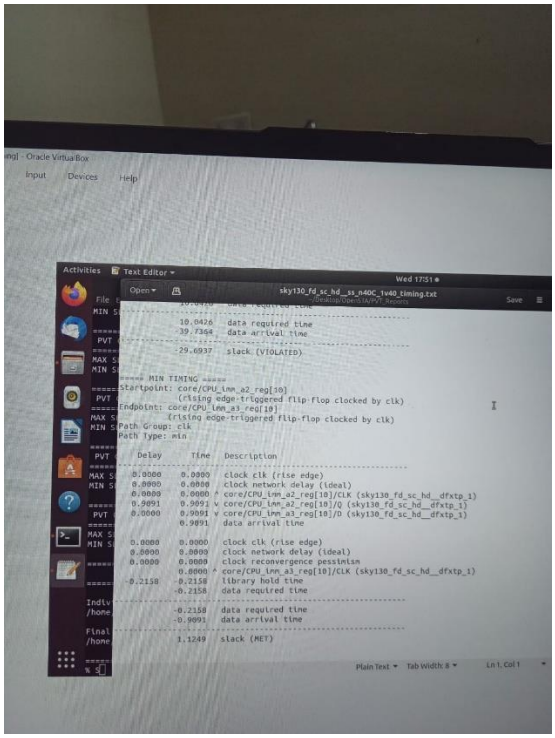
- **sky130_fd_sc_hd__ss_n40C_1v35**



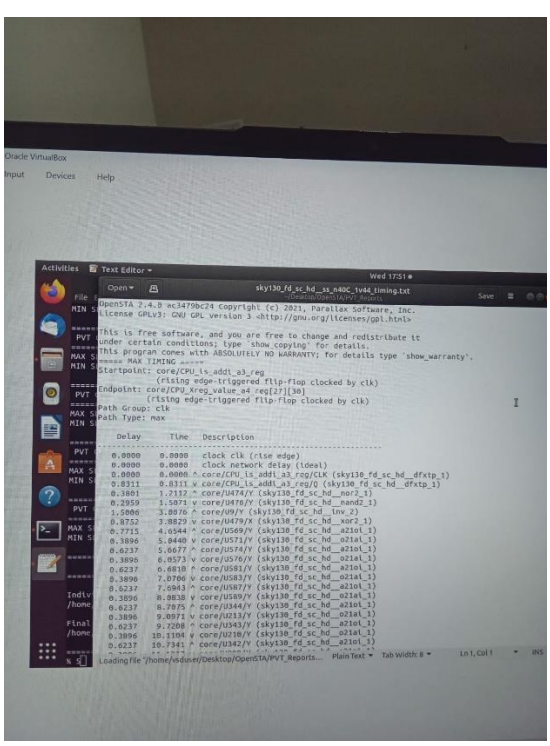


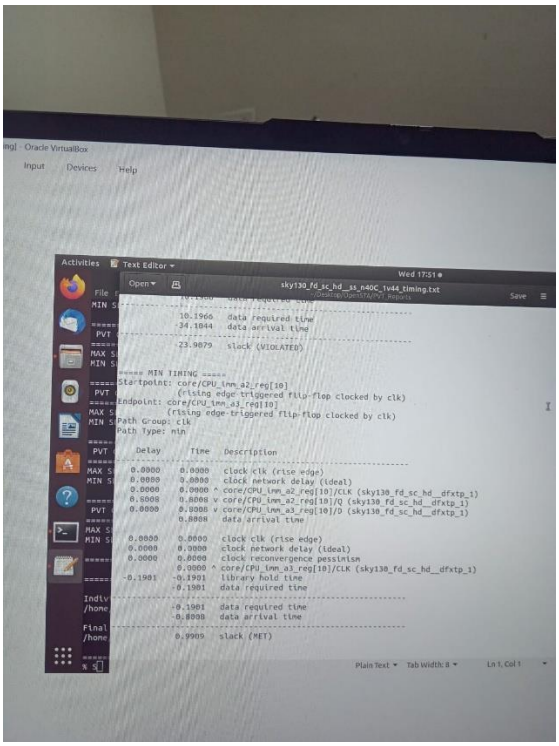
- **sky130_fd_sc_hd__ss_n40C_1v40**



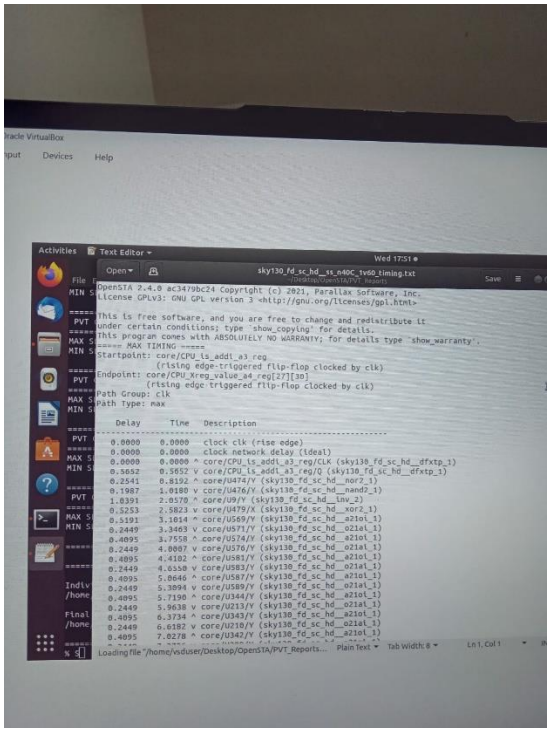


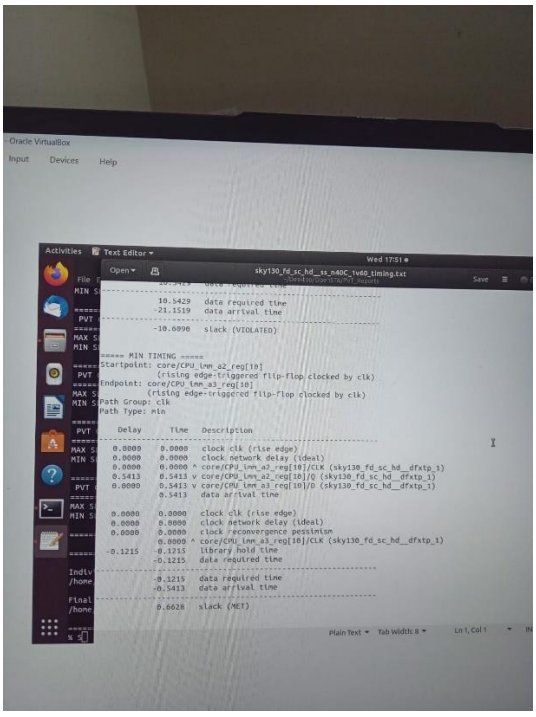
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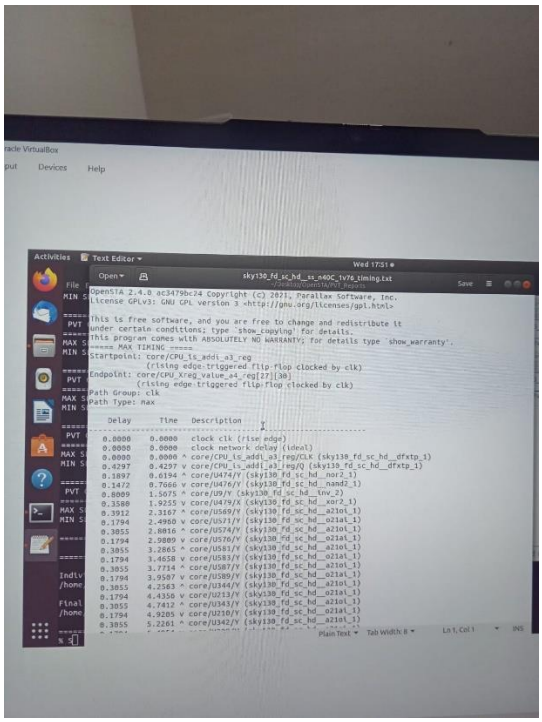


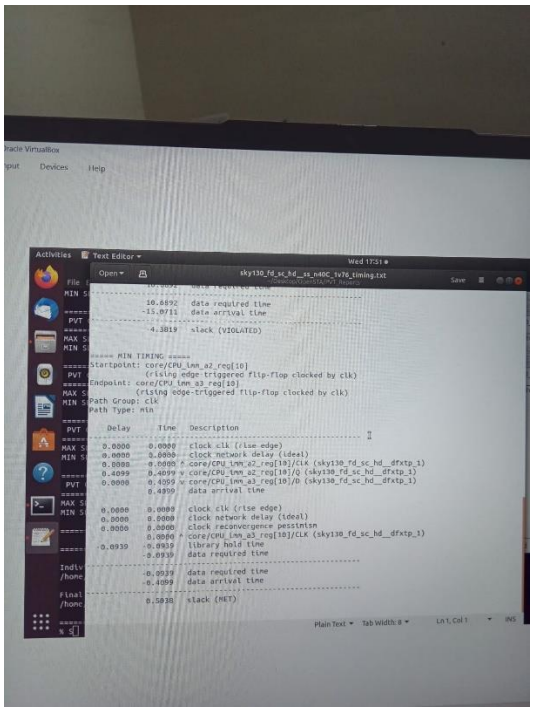
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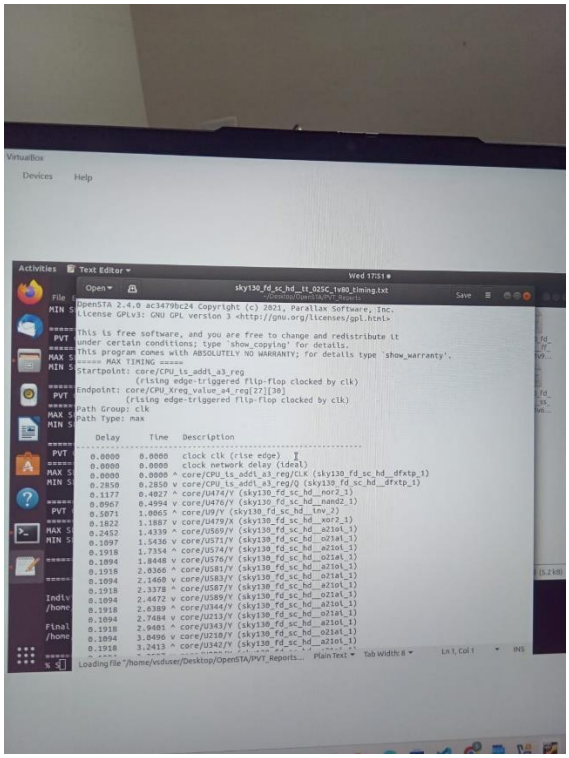


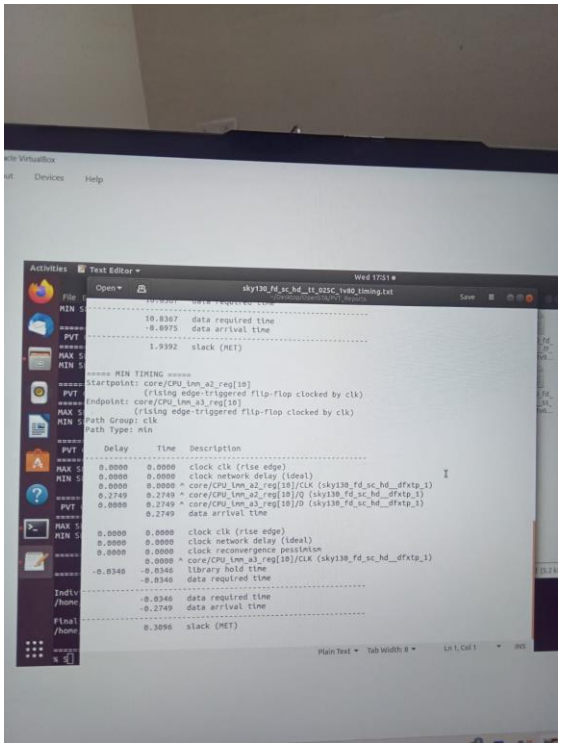
- **sky130_fd_sc_hd__ss_n40C_1v76**



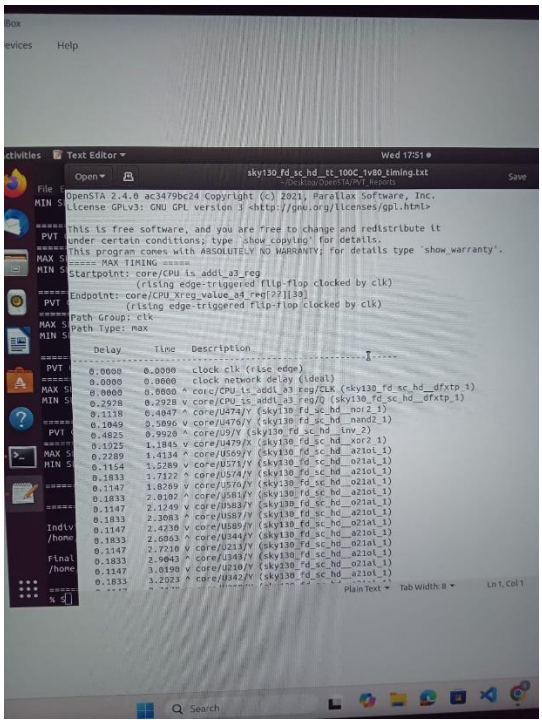


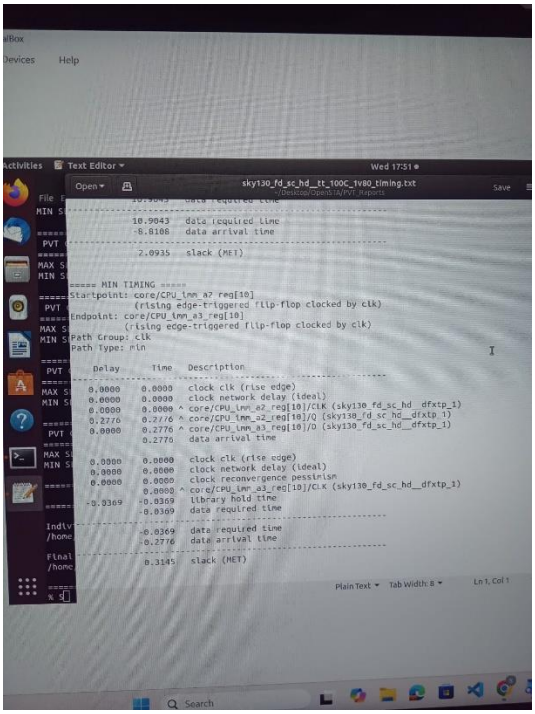
- **sky130_fd_sc_hd__tt_025C_1v80**





- **sky130_fd_sc_hd__tt_100C_1v80**





8. Key Observations

• The maximum/setup slack varies strongly across the 16 PVT corners.
• The collected maximum slack ranges from −62.5632 ns to +5.5919 ns.
• The collected minimum/hold slack is positive for all 16 corners, ranging from 0.1875 ns to 1.8296 ns.
• The TT 25°C, 1.80 V corner reports 1.9392 ns maximum slack and 0.3096 ns minimum slack.
• The slow-slow, low-voltage corners show the largest negative maximum slack values in this analysis.
• The FF corners show positive maximum slack across the listed voltage and temperature conditions.

9. Conclusion

Static timing analysis was performed for all 16 complete Sky130 HD timing-library corners available in the timing directory. Both maximum/setup and minimum/hold timing results were collected and compared. The resulting table and graphs provide a PVT-wise view of timing behavior and identify the corners with negative maximum slack for further timing optimization.